Amit Kumar

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Objective

A postdoctoral researcher with expertise in developing biomimetic, responsive soft materials for biomedical and robotic applications. Skilled in interdisciplinary collaboration, advanced material characterization, and translational research. Dedicated to advancing technologies at the interface of adaptive robotics and biomedical science.

Research Area

My research area includes:

- **Stimuli-Responsive Systems:** pH-responsive and solvent-adaptive hydrogels for targeted drug delivery.
- **Biomaterials Design:** Engineering dynamic biomaterials for medical applications, including anti-fouling and vascular graft systems.
- **Biomimetics and Soft Robotics:** Nature-inspired actuation systems for biointegrated devices and soft robotics.
- **Advanced Manufacturing:** Developing multifunctional materials using 3D and 4D printing techniques.

Education

- **Master of Science(M.S.) + Doctor of Philosophy (Ph.D.)- 9.75(CGPA)** July'17 - July'24

  Dual Degree Interdisciplinary Research Program,
  Department of Biotechnology, Indian Institute of Technology Madras, Chennai, India.

  Thesis: *Harnessing Solvent-Responsive Actuation Behavior of Polymers for Biomedical Applications*

  Supervisors: *Prof. Nitish R. Mahapatra, Department of Biotechnology,*
  Prof. Pijush Ghosh, Department of Applied Mechanics & Biomedical Engineering.

- **Bachelor of Technology(B.Tech.)- 70.42(CPI)** Aug'13 - May'17

  Department of Biotechnology, Delhi Technological University, Delhi, India.

  Major Project: *Review report on biosensors for detection of plant toxins.*

  Supervisor: *Prof. B. D. Malhotra, Department of Biotechnology.*

Professional Experience

- **Postdoctoral Researcher**, Istituto Italiano di Tecnologia(IIT), Italy 1st June'25 - Present

  (Mentor: Prof. Barbara Mazzolai, Director, Bioinspired Soft Robotics (BSR) Laboratory, IIT, Genoa)

  - Working on plant-inspired soft robotics for soil exploration applications
  - Designing biodegradable hydrogel–elastomer composite actuators that can move through the soil while enabling controlled release of bioactive molecules (e.g., nutrients, antimicrobial agents) in response to environmental cues—supporting sustainable soil health management.
  - Integration of luminescence-based and colorimetric sensing strategies for real-time monitoring of soil parameters such as pH, temperature, UV exposure, and harmful biomolecules

- **Research Associate**, IITM, Chennai, India 19th Sept'24 - 28th Feb'25

  (Mentor: Prof. Pijush Ghosh, Nanomechanics Lab, Department of Applied Mechanics & Biomedical Engineering, IITM, Chennai)

  - Developed dual(solvent and light) responsive bilayer polymer films for regulating biofilm removal.
  - Effective biofilm removal was tested using actuation-induced curvature of dual-responsive films by cyclic folding/unfolding.
  - Designed and Developed 3D printed catheter model via FDM printing.

- **Industrial Trainee**, National Physical Laboratory, CSIR, Delhi, India — Jun'16 - Aug'16

(Mentor: Dr. G. Sumana Gajjala, Biomolecular Electronics and Conducting Polymer Research Group, NPL, CSIR)

  - Worked on FTIR, Zeta-Sizer, Electrophoretic Deposition Technique, And Electrochemical Analyzer.
  - Used Langmuir-Blodgett Film Deposition Technology for making a Monolayer of Molybdenum Di-Sulphide and Gold Nanoparticles (AuNPs) composite.

- **Industrial Trainee**, INMAS, DRDO, Delhi, India — Dec'15 - Jan'16

(Mentor: Dr. Aseem Bhatnagar, Department of Nuclear Medicine, INMAS, DRDO)

  - Assisted in making Melatonin-Caffeine Tablets for clinical trials that act as Radio-Protector.
  - Enhanced probiotic curd formation rate by using skim milk powder, calcium chloride, pectin, lactose, and lactobacillus spores
  - Learned about working of tablet punching machine, disintegration tester, radiolabelling with technisium, thin layer chromatography, gamma scintigraphy scan, CAPRAC-t, UV-IR spectrometer.

Research Projects

- **Efficient Reduction of the Scrolling of Descemet Membrane Endothelial Keratoplasty (DMEK) Grafts by Engineering the Medium**
  - The diffusion-coupled deformation phenomenon based mechanism for an out-of-plane folding or scrolling of the graft tissue was explained.
  - Chitosan bilayer-based experimental models developed to mimic DMEK graft scrolling.
  - Designed solvent characteristics to reduce scroll tightness.

- **Engineering the Nonmorphing Point of Actuation for Controlled Drug Release by Hydrogel Bilayer across the pH Spectrum**
  - Developed a biocompatible pH-responsive bilayer film consisting of Chitosan (CS) and Carboxymethyl Cellulose (CMC) cross-linked via Citric Acid with tunable non-morphing points.
  - Bidirectional actuation induced controlled release of drug molecules were illustrated.
  - The structural geometries of the bilayer film were engineered for nature-inspired actuation patterns.

- **Bidirectional Actuation Controlled Multidrug Release via pH-Responsive Bilayer for Tumor Therapy**
  - pH-responsive actuation behaviour of CS/CMC films were tuned by varying different solvent characteristics(acid-base combination).
  - Curvature induced drug release kinetics were studied for multidrug(Cisplatin, 5-fluorouracil, Quercetin) loaded pH-responsive film.
  - In-vitro experiments were conducted on MDA-MB 231 tumor cells lines.

- **Superhydrophobic Asymmetric pH-responsive Soft Actuators: Implications for the Development of Anti-fouling Medical Devices**
  - Developed an asymmetric Chitosan film with both superhydrophobic top and hydrophilic bottom surface through the self-assembly of Methyl Trichloro Silane (MTS) modified silica nanoparticles.
  - The asymmetric films with single active material i.e. Chitosan, showed bidirectional pH-responsive behavior.
  - In-vitro antibacterial adhesion, antibacterial activity and bacterial infiltration with *E. coli* (DH5α) and *S. Aureus* (RN4220) cells studies were conducted.
  - 3D printed architectures were actuated to obtain structures with significance in biomedical applications.

- **Role of Solvent- and Light- Responsive Film in Regulating Bio-film Formation**
  - Developed a bilayer of hydrophobic Polydimethylsiloxane (with light responsive dye)/hydrophilic Chitosan (solvent responsive) via modified silica nanoparticles at the interface.
  - Effect of actuation induced curvature on adhesion and growth of *E. coli* (DH5α) and *S. Aureus* (RN4220) cells were studied.
  - Significant bacterial biofilms were removed via repeated cycles of actuation.

Research Publications

1. **Amit Kumar**[1], Shaik Sameer Basha[1], Mukilarasi B, Vimalraj Selvaraj, Pijush Ghosh, Swathi Sudhakar. Leveraging cancer microenvironment pH shift for actuation-induced multidrug release from chitosan/carboxymethyl cellulose bilayer film. *International Journal of Biological Macromolecules*, **(2025)**, 144701. DOI: 10.1016/j.ijbiomac.2025.144701 **(Impact Factor- 8.7)**

2. **Amit Kumar**, Smruti Parimita, Kumari Kiran, Nitish R. Mahapatra, Pijush Ghosh. Superhydrophobic Asymmetric pH-responsive Soft Actuators: Implications for the Development of Anti-fouling Medical Devices. *Chemical Engineering Journal*, **(2024)**, 154772. DOI: 10.1016/j.cej.2024.154772 **(Impact Factor- 13.5)**

3. Nidhi Gupta[1], **Amit Kumar**[1], Pravin K Vaddavalli, Nitish R. Mahapatra, Akhil Varshney, Pijush Ghosh. Efficient Reduction of the Scrolling of Descemet Membrane Endothelial Keratoplasty Grafts by Engineering the Medium. *Acta Biomaterialia*, **(2023)**, 171, 239-248. DOI: 10.1016/j.actbio.2023.09.024 **(Impact Factor- 10.3)**

4. **Amit Kumar**, Raja Rajamanickam, Joyita Hazra, Nitish R. Mahapatra, Pijush Ghosh. Engineering the Nonmorphing Point of Actuation for Controlled Drug Release by Hydrogel Bilayer across the pH Spectrum. *ACS Applied Materials & Interfaces*, **(2022)**, 14, 50, 56321–56330. DOI: 10.1021/acsami.2c16658 **(Impact Factor- 8.5)**

5. Smruti Parimita, **Amit Kumar**, Hariharan Krishnaswamy, Pijush Ghosh. 4D printing of pH-responsive bilayer with programmable shape-shifting behaviour. *ACS European Polymer Journal*. **(2024)**. DOI: 10.1016/j.eurpolymj.2024.113581 **(Impact Factor- 6.0)**

6. Smruti Parimita, **Amit Kumar**, Hariharan Krishnaswamy Pijush Ghosh. Solvent triggered shape morphism of 4D printed hydrogels. *Journal of Manufacturing Processes*, **(2023)**, 85, 875-884. DOI: 10.1016/j.jmapro.2022.11.065 **(Impact Factor- 7.0)**

7. Anas Saifi, Smruti Parimita, **Amit Kumar**, Pijush Ghosh. Programmable 4D-Printed Soft Actuators: Harnessing Bending Strain Distribution for Embedded Topological Functionality. *ACS Applied Engineering Materials*. **(2025)**. DOI: 10.1021/acsaenm.5c00225 **(Impact Factor- 3.5)**

Technical Skills

- **Laboratory Proficiencies:**
  - **Responsive biomaterials fabrication:** hydrogels, bilayers, 3D and 4D printed structures
  - **Advanced microscopy and spectroscopy:** SEM, FT-IR, UV-Vis
  - **Cellular assays:** MTT, live/dead bacterial viability, antibacterial studies
- **Computational and Analytical Tools:**
  - **Visualization & Design tools:** Origin, Graph Prism, Chemdraw, Inkscape, Blender, Filmora 9.0, TinkerCad, Ultimaker Cura
  - **Documentation:** Microsoft Office, LaTeX

Posters and Presentations

1. **Amit Kumar**, Smruti Parimita, Kumari Kiran, Nitish R. Mahapatra and Pijush Ghosh,"Superhydrophobic Asymmetric Solvent Responsive Actuator for Antifouling Medical Devices", *Applied Mechanics and Biomedical Engineering (AMBE) Research Palooza*, 5th July 2024, IIT Madras, Chennai, India.

2. **Amit Kumar**, Nidhi Gupta, Nitish R. Mahapatra and Pijush Ghosh, "Strategies to attain underwater actuation of solvent-responsive chitosan and its biomedical applications", *33rd Annual Conference of the European Society for Biomaterials* (ESB-2023), 4th – 8th September 2023, Davos, Switzerland.

3. **Amit Kumar**, V.P. Anju, Nitish R. Mahapatra and Pijush Ghosh, "Impact of Tuning Solvent Characteristics on Self-folding of Biopolymer Thin Film", *Active Polymeric Materials and Microsystems* (APMM-2019), 16th – 19th September 2019, Dresden, Germany.

4. **Amit Kumar**, Smruti Parimita, Nitish R. Mahapatra and Pijush Ghosh, "pH-Responsive Soft Actuator for Biomedical Applications", *3D Graphy Engineering and Medical 2023*, (3DGRAPHY- 2023), 9th - 10th December 2023, IIT Bombay, Maharashtra, India.

5. **Amit Kumar**, Nitish R. Mahapatra and Pijush Ghosh, "pH-responsive bilayer films with engineered non-morphing point of actuation for controlled model drug release", *14th International Conference on Advancements in Polymeric Materials* (APM-2023), 17th – 19th March 2023, CIPET (SARP)-APDDRL, India.

Workshop Attended

1. Additive Manufacturing of Bio-Implants - Academic & Industry Perspectives, 10^{th} - 11^{th} March 2023, Indian Institute of Technology Madras, India.

Research Awards and Grants

- **Marie Curie Postdoctoral Fellowship- Seal of Excellence** 2026
  - The proposal "4DEX-mplant" secured 90.80% was recognised as a high-quality project proposal by European Commission.
- **Research Excellence Award-** IIT Madras 2024
  - Best Research Publication in the Department of Biotechnology, awarded by IIT Madras during Dec 2023- Dec 2024.
- **Institute Research Award-** IIT Madras 2024
  - Best Doctoral Research in the Department of Biotechnology, awarded by IIT Madras.
- **First Prize-** Sci-Tech Exposium, IIT Madras 2024
  - First prize in oral presentation at Applied Mechanics and Biomedical Engineering (AMBE) Research Palooza.
- **Graduate Aptitude Test in Engineering (GATE)-** India 2017
  - National level scholarship exam to pursue M.S. (by research) from IIT Madras.

Position of Responsibility

- **Workshop Organiser- ieee ras ROBOSOFT 2026**, Kanazawa, Japan April'26
  - Title: Functional and Untethered Soft Machines Powered by the Environment
  - Theme of the workshop: Can soft machines achieve true autonomy by dynamically adapting to environmental changes or by harvesting and utilizing ambient energy sources?
  - The workshop includes: Invited talks, A concept competition, A demonstration competition and A poster competition.
- **Teaching Assistant- Genetic Engineering Lab**, IIT Madras Aug'19 - Dec'19
  - Conducted competant cells, transformation, plasmid isolation, gel electrophoresis, restriction digestion and PCR amplification experiments.
- **Teaching assistant- Microbiology Lab**, IIT Madras Aug'18 - Dec'18
  - Successfully conducted experiments for media preparation, pure culture isolation, gram staining, and selective media preparation.

Position of Responsibility(Non-Academic)

- **Hostel Football Captain- Mahanadhi Hostel**, IIT Madras Aug'22 - May'23
  - Led the Postgraduates Hostel team to become Champions in 11 v 11 football tournament during Schroeter, an Annual Inter Hostel Sports Meet 2022-23 among 16 hostels.
- **Institute Athletics Captain- IIT Madras** Aug'22 - May'23
  - Institute Athletics team finished 2nd place in annual Inter IIT Sports Meet 2022-23 among 23 IITs.

Extracurricular Activities

- Secured 3rd place in both football and Pole vault (athletics) at Inter IIT Sports Meet, IIT Gandhinagar in December, 2023.
- Captained hostel football team to become Champions in 11v11 football tournament during Schroeter, an Annual Inter Hostel Sports Meet 2022-23 among 16 hostels.
- Held Inter IIT Meet Record holder for 3 years from 2018-2022 in Pole Vault with a height of 3.82m and won the Gold Medals for three consecutive year in Inter IIT Sports Meet among 21 IITs across India.
- Bagged 1st place in 8km Road Race, Dean's trophy 2017-18 held at IIT Madras.

Academic Referees

- Prof. Barbara Mazzolai, Director, Bioinspired Soft Robotics Group, Italiano di Tecnologia (IIT), Genoa, Italy (email: barbara.mazzolai@iit.it)
- Prof. Nitish R. Mahapatra, Professor, Department of Biotechnology, Bhupat and Jyoti Mehta School of Biosciences, Cardiovascular Genetics Lab, Indian Institute of Technology Madras, India (email: nmahapatra@iitm.ac.in)
- Prof. Pijush Ghosh, Professor, Department of Applied Mechanics & Biomedical Engineering, Nanomechanics Lab, Indian Institute of Technology Madras, India (email: pijush@iitm.ac.in)
- Dr. Nidhi Gupta, Senior Consultant, Dr. Shroff's Charity Eye Hospital, India (email: nidhiophthal@gmail.com)